

Assignment 3

1) A bit stream is 10101010 is transmitted using the standard CRC method. The generator polynomial is $x^3 + 1$. Determine the transmitted bit string.

→ $1x^3 + 0x^2 + 0x + 1x^0$
1 0 0 1

$$\begin{array}{r}
 1001 \overline{) 10101010000} \\
 \underline{\oplus 1001} \\
 001110 \\
 \underline{\oplus 1001} \\
 01111 \\
 \underline{\oplus 1001} \\
 01100 \\
 \underline{\oplus 1001} \\
 01010 \\
 \underline{\oplus 1001} \\
 001100 \\
 \underline{\oplus 1001} \\
 0101
 \end{array}$$

101

∴ The transmitted string is 101010101011

2) using the standard CRC method, a bit stream 11011011 is transmitted. The generator polynomial is $x^4 + x^2 + 1$. Find original bit string sent:-

→ $x^4 + x^2 + 1$
→ $1x^4 + 0x^3 + 1x^2 + 0x^1 + 1x^0$
1 0 1 0 1

$$\begin{array}{r}
 10101 \overline{) 1101101100000} \\
 \underline{\oplus 10101} \downarrow \\
 011100 \downarrow \\
 \underline{\oplus 10101} \downarrow \\
 010011 \downarrow \\
 \underline{\oplus 10101} \downarrow \\
 0011000 \downarrow \\
 \underline{\oplus 10101} \downarrow \\
 011010 \downarrow \\
 \underline{\oplus 10101} \downarrow \\
 011110 \downarrow \\
 \underline{\oplus 10101} \downarrow \\
 01011
 \end{array}$$

original bit string is .

11011011, 1011 //

3. If a bit stream 11110000. The generator polynomial is $x^3 + x^2 + 0x + 1$, what is the transmitted bit string.

→ 11110000 $x^3 + x^2 + 0x + 1$
 1 1 0 1

$$\begin{array}{r}
 1101 \overline{) 111100000000} \\
 \underline{1101} \\
 0010000 \\
 \underline{1101} \\
 01010 \\
 \underline{1101} \\
 01110 \\
 \underline{1101} \\
 001100 \\
 \underline{1101} \\
 00010
 \end{array}$$

∴ Transmitted bit is-

1111000001011

9) Transmitting a bit stream 1010101010 using standard CRC method with the generator polynomial $x^5 + x^4 + x^2 + 1$. What is actual transmitted bit string.

→ 1010101010 $x^5 + x^4 + x^2 + 1$
110101

$$\begin{array}{r}
 110101 \overline{) 1010101010000000} \\
 \underline{110101} \\
 0111111 \\
 \underline{110101} \\
 00101001 \\
 \underline{110101} \\
 0111000 \\
 \underline{110101} \\
 00110100 \\
 \underline{110101} \\
 000001000
 \end{array}$$

∴ original transmitted bit is.

10101010100100011

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dhulikhel, Kavre.

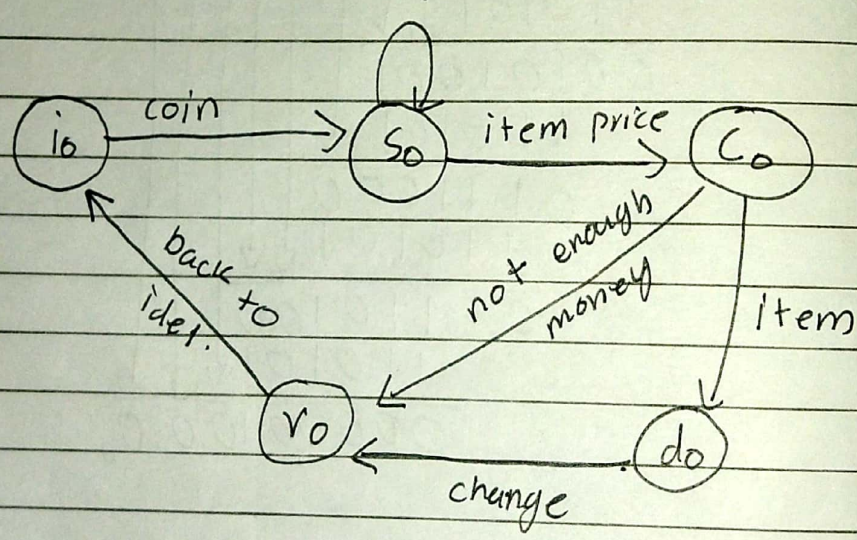
Embedded Systems (comp 306) Assignment - 3

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DOCSE

Develop a Finite state Machine (FSM) that the behaviour of a simple vending machine including states for accepting coins, selecting product and dispensing item.

states	transition functions
idel (i ₀)	coin, no item selected
coin count (c ₀)	item price, item.
dispence (d ₀)	not enough money
return money (r ₀)	Change, idel.
select (s ₀)	no item selected



Present state	Input	Nent state.
i ₀	coin	s ₀
s ₀	no item selected	s ₀
s ₀	price	c ₀
c ₀	item	d ₀
c ₀	not enough money	r ₀ .
d ₀	change	r ₀
r ₀	back to idel	i ₀ .

2) create a finite state Machine (FSM) to simulate the behaviour of basic elevator system, including states for moving up/down and opening/closing doors.

→ states.

idle (i₀)

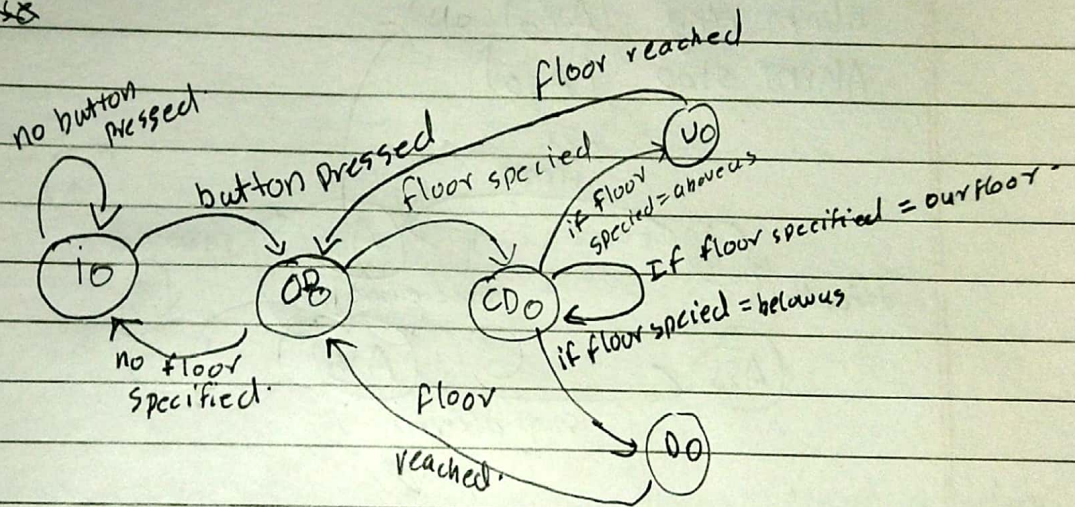
Open door (O_{D0})

up (U₀)

close door (C_{D0})

down (D₀)

~~close~~



Present state	Input	Next state
i ₀	No buttons pressed	i ₀
i ₀	floor button pressed	OD ₀
OD ₀	floor specified	CD ₀
CD ₀	if floor specified = above us	U ₀
CD ₀	if floor specified = below us	D ₀
CD ₀	if floor specified = our floor	CD ₀
U ₀	floor reached	OD ₀
D ₀	floor reached	OD ₀
OD ₀	no floor specified	i ₀

3) Design a finite state machine (FSM) that represents the behaviour of simple alarm clock, including states for setting the time, setting alarm, & triggering alarm.

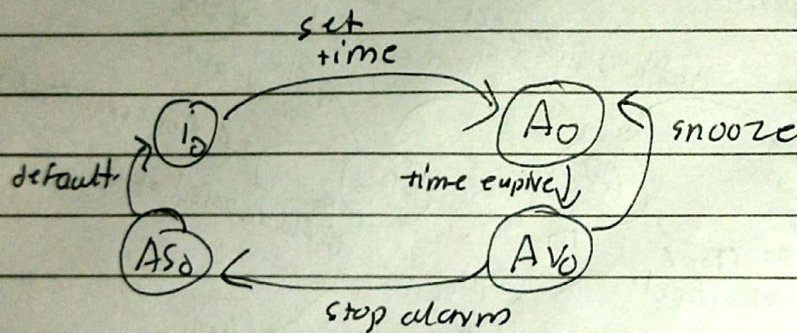
→ States

Idle (i_0)

Alarm set (A_0)

Alarm ring (A_{r0})

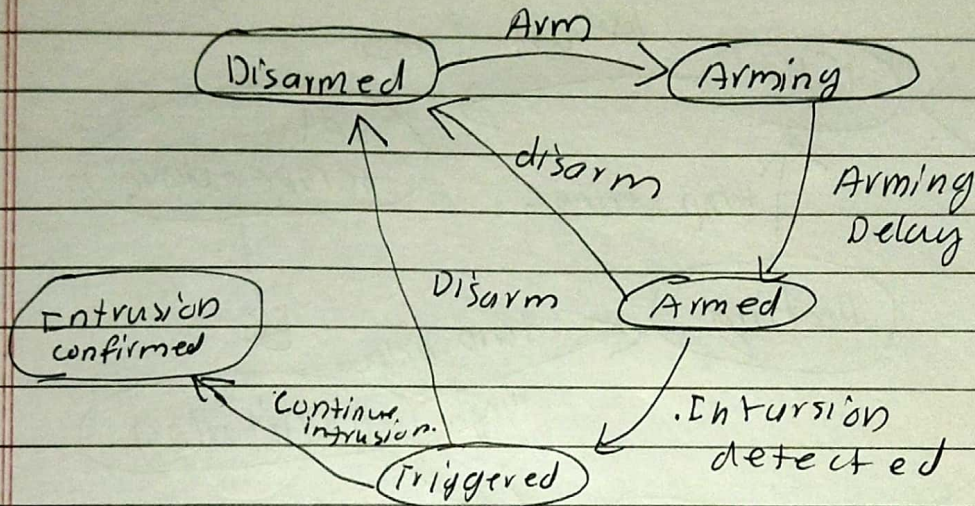
Alarm stop (A_{s0})



Present state	Next Input	Next state
i_0	set time	A_0
A_0	time expire	A_{r0}
A_{r0}	stop alarm	A_{s0}
A_{s0}	default	i_0
A_{r0}	snooze	A_0

4) Develop a finite state machine (FSM) to control the operation of a basic home security system, including state for arming / disarming and detecting intrusions

→ states.



Present state	Input	Next state
Disarm	Arm	Arming
Arming	Arming delay	Armed
Armed	Intrusion detected	Triggered
Triggered	continue intrusion	Intrusion confirmed
Triggered	disarm	Disarmed
Armed	Disarm	Disarmed

5) create a FSM to model the behavior of digital thermostat; including states for setting the temperature & adjusting heating/cooling system.

